MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION
APRIL 2026 : SECOND SEMESTER
Discipline Specific Core Course (DSC) - MG2DSCAQC100
ORNAMENTALFISH CULTURE AND BREEDING
(2024 ADMISSION ONWARDS)
Duration: 1 Hour 30 Minutes
Maximum Marks: 50
Q paper Code : 25901597

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Fill in the Blanks/ Name the Following
Answer all questions
Each question carries 1 marks

1. Which ornamental fish species is famous for its bright, long fins? [U] / [CO2]
Betta Fish (*Betta splendens*)
2. Which marine ornamental fish is known for its symbiotic relationship with sea anemones?. [K] / [CO2]
Clown fish (*Amphiprion ocellaris*)
3. What is considered the second-largest hobby in the world after photography? [K] / [CO4]
Ornamental fish culture
4. Mention any one benefit of ornamental fish keeping. [K] / [CO1]
Income generation and employment, Low investment, high returns, Export potential, Stress relief and aesthetic value, Educational and research value, Eco-friendly activity, Promotes entrepreneurship, Conservation etc. Any one benefit
5. Name two types of artificial fish feeds based on their water content. [K] / [CO3]
Wet feed and dry feed
5. Scientific name of Eel grass. [K] / [CO1]
Zostera marina
6. Write the scientific name of any one marine ornamental shrimp of India. [K] / [CO2]
*Lysmata amboinensis, Lysmata debelius, Stenopus hispidus, Periclimenes brevicarpalis, Periclimenes pedersoni (*any one)
7. Why is starvation done before transporting ornamental fish? [U] / [CO4]
Reduces waste production, Maintains water quality, Prevents oxygen depletion
Minimizes stress and mortality
9.Which category of fish lays eggs instead of giving birth to live young? [K] / [CO4]
Oviparous fishes
10.Name any two funding agencies involved in promoting sustainable ornamental fish farming [K] / [CO5]
National Fisheries Development Board (NFDB), Marine Products Export Development Authority (MPEDA), National Bank for Agriculture and Rural Development (NABARD), Ministry of Fisheries Animal Husbandry and Dairying, Department of Fisheries (any one)

Part B
Short Answer Questions
Answer any 6 questions
Each question carries 5 marks

11. Explain how an undergravel filter works. Why is it considered less effective in modern aquariums? [U] / [CO1]

An undergravel filter (UGF) is a biological filtration system placed beneath the aquarium substrate (gravel). Uplift tubes, Air pumps or powerheads, suction, pulling water down through the gravel, beneficial bacteria colonize the substrate, *Nitrosomonas, Nitrobacter* These bacteria convert toxic ammonia → nitrite → nitrate (biological filtration), Filtered water then rises up through the uplift tubes

Less effective in modern aquariums : Accumulation of waste, Difficult maintenance, Limited filtration efficiency, Not suitable for planted tanks, Inflexibility, Better alternatives available (canister filter etc.)

12. Compare the setup, maintenance, and challenges of freshwater and marine aquariums. [U] / [CO2]

1. Setup

Freshwater Aquarium : Simple, beginner-friendly, Uses dechlorinated tap water, Basic equipment: tank, filter, heater, light, Low cost and easy availability of fish like Guppy and Goldfish

Marine Aquarium: Complex setup, Requires saltwater mix and precise salinity control, Additional equipment: protein skimmer, hydrometer/refractometer, live rock, High initial cost; species like Clownfish are popular

2. Maintenance

Freshwater Aquarium: Easier water changes and cleaning, Less strict monitoring of water chemistry, Fish are generally more tolerant to minor fluctuations

Marine Aquarium: Requires regular monitoring of salinity, pH, calcium, alkalinity, Frequent and careful water changes with prepared saltwater, More time-consuming and technically demanding

3. Challenges

Freshwater Aquarium: Algae growth and occasional disease issues, Overstocking and poor filtration can cause problems, Generally manageable for beginners

Marine Aquarium: Sensitive organisms; small changes can be harmful, Maintaining stable water chemistry is difficult, Higher risk of losses and higher cost of livestock, Requires good knowledge and experience

13. Explain the breeding behaviour of a viviparous ornamental fish. [U] / [CO4]

Viviparous ornamental fishes, live-bearers, Mention any one of the common example such as guppy, molly for its breeding behaviour

1. Courtship

2. Mating (internal fertilization) : Male, gonopodium, sperm transfer

3. Gestation period: The fertilized eggs develop within the female, Gestation usually lasts about 3–4 weeks (varies with species and temperature)

4. Development of young: Embryos receive nourishment from yolk (in most species), The female's abdomen enlarges and a gravid spot becomes visible

5. Parturition (birth): The female gives birth to fully formed, free-swimming fry, Each brood may contain 20–100 or more young

6. Post-birth behaviour: No parental care, Adults may eat the fry, so separation is needed in aquariums

14. How egg laying ornamental fishes are categorised? [K] / [CO4]

Egg-laying ornamental fishes (oviparous fishes) are categorized based on their spawning habits and the nature of their eggs:

Explain terms such as 1. Egg scatterers, 2. Egg depositors, 3. Bubble nest builders, 4. Mouth brooders, 5. Nest builders (non-bubble) with examples

15. Distinguish between Ligulosis and Bucephalus diseases in fishes. [K] / [CO2]

1. Causative organism

Ligulosis : Caused by the larval stage (plerocercoid) of tapeworms of the genus Ligula

Bucephalus disease: Caused by trematode parasites of the genus Bucephalus

2. Site of infection

Ligulosis: Parasites are found in the body cavity (abdomen)

Bucephalus disease: Parasites mainly infect the gonads (reproductive organs)

3. Symptoms

Ligulosis: Distended abdomen, Emaciation , Abnormal swimming , Reduced growth

Bucephalus disease: Destruction of gonadal tissue, Sterility in fish, Reduced reproduction

4. Effect on fish

Ligulosis : Causes mechanical pressure on internal organs, leading to poor health and sometimes death

Bucephalus disease: Causes parasitic castration, affecting breeding capacity rather than immediate survival

5. Life cycle involvement

Ligulosis: Involves fish and fish-eating birds as hosts

Bucephalus disease : Involves snails as intermediate hosts and fish as secondary hosts

16. Explain the steps in induced breeding. [K] / [CO1]

Explain steps such as 1. Selection of broodstock, 2. Conditioning of brooders, 3. Hormone injection, 4. Spawning (breeding), 5. Fertilization, 6. Incubation of eggs, 7. Hatching, 8. Larval rearing

17. Discuss the importance of salinity and pH balance in a marine aquarium and how they are maintained. [U] / [CO2]

Maintaining salinity and pH balance is critical in a marine aquarium because marine organisms like Clownfish and corals are highly sensitive to water chemistry.

1. Importance of Salinity: Controls osmoregulation (water–salt balance), Affects metabolism, growth, and survival, Essential for invertebrates (corals, shrimps) for proper physiological functions, Sudden changes can cause stress or death, Optimal range: 30–35 ppt

2. Importance of pH : Influences biological processes and enzyme activity , Affects toxicity of ammonia, Essential for calcification in corals and shell-forming organisms, Sudden fluctuations cause stress and disease, Optimal range: 8.0–8.4

3. Maintenance of Salinity: Use marine salt mix with purified water, Measure regularly with hydrometer or refractometer, Compensate evaporation by adding freshwater, Perform regular water changes with properly mixed saltwater

4. Maintenance of pH: Regular water changes to maintain stability, Maintain proper alkalinity (buffering capacity), Use buffers or conditioners if needed, Ensure good aeration and filtration, Avoid overfeeding and waste accumulation

18. Describe global trades and market trends related to ornamental fish industry [K] / [CO5]

The ornamental fish industry, rapidly growing global sector within aquaculture, involving the trade of live aquarium fishes and related products.

1. Global trade overview

The ornamental fish industry is a multi-billion-dollar global business, valued at about USD 5.8 billion in 2022 and expected to reach over USD 11 billion by 2030

More than 2,500 freshwater species and 800 marine species are traded worldwide

The trade involves a complex value chain: collectors → breeders → exporters → wholesalers → retailers → hobbyists

2. Major exporting and importing regions

Exporting regions : Asia-Pacific dominates production (about 50%+ share)

Major exporters: China, Singapore, Thailand, Indonesia, India

Tropical climate favors low-cost breeding and high species diversity

Importing regions : USA, Europe, and Japan are major consumers

The U.S. alone represents a large share of global demand, especially for marine fish

3. Market trends : Increasing demand for aquarium fish, Dominance of freshwater species, Growth of commercial sector, Shift towards captive breeding, Expansion of online and global trade

4. Emerging trend: Aquascaping and designer aquariums, Use of advanced aquarium technologies (automated systems), Rising interest in rare and exotic species, Increasing focus on eco-friendly and sustainable practices

5. Challenges in global trade : Heavy dependence on wild-caught marine fish in some sectors, Regulatory restrictions and biodiversity concerns, Risk of disease transmission during transport, Lack of standardization and traceability in supply chains

6. Indian scenario: India has high potential due to biodiversity and climate, Still contributes a small share to global trade but growing rapidly

19. Describe the three main types of filtration used in aquariums with examples. [K] / [CO1]

Mechanical filtration: Function: Removes solid particles like uneaten food, fish waste, and debris from water, Water passes through filter media that physically traps particles. Examples:Sponge filters, Filter pads/floss in internal or external filters

2. Biological filtration: Function: Removes toxic nitrogenous wastes through beneficial bacteria.Bacteria convert ammonia → nitrite → nitrate (nitrogen cycle) Occurs on surfaces like gravel, sponges, and bio-media

Examples:Undergravel filter systems, Bio-rings and ceramic media in canister filters

Fish like Goldfish produce significant waste, making biological filtration essential.

3. Chemical filtration: Function:Removes dissolved impurities, toxins, odors, and discoloration.Uses chemical media that adsorb or neutralize substances.Examples:Activated carbon, Zeolite, Resins

Part C
Essay
Answer any 1 questions
Each question carries 10 marks

20. Write an essay on the broodstock management, breeding behaviour and rearing of juveniles of an ornamental fish which shows parental care. [K] / [CO4]

Explain the broodstock management, breeding behaviour and rearing of Gourami/fighter fish

1. Broodstock Management : Selection of brooders (size, health, ration etc), Sex differentiation, Feeding, Water quality maintenance
2. Understanding of breeding behaviour and parental care, bubble nest builders

3. Breeding tank setup according to the breeding behaviour and parental care
4. Hatching

5. Rearing of Juveniles : Larval stage, Free-swimming stage, Feeding, Water management

21. Write an essay on the different propagation methods of ornamental plants [K] / [CO1]

Propagation of ornamental plants is the process of producing new plants from existing ones. Propagation methods : sexual (seed) and asexual (vegetative) methods.

1. Sexual propagation (by seeds), Advantages, Disadvantages

2. Asexual (vegetative) propagation

This method produces plants genetically identical to the parent plant. Explain methods such as stem cutting, leaf cutting, plantlets, offsets shoots, shooting or sprouting, Layering, Grafting, Budding, runner division, Specialized vegetative structures such as Bulbs, Corms, Rhizomes, Tubers , Micropropagation (tissue culture)

22. Discuss the diversity and distribution of freshwater indigenous ornamental fishes in India. [K] / [CO2]
India is one of the world's richest regions in terms of freshwater fish biodiversity.

1. Diversity of indigenous ornamental fishes : 300+ species of freshwater ornamental fishes, many of which are endemic

Belong to important groups such as: Cyprinids (carps and minnows), Danionins (barbs and danios), Catfishes, Cichlids , Loaches and gobies, mention examples

Many species are small-sized, making them ideal for aquarium keeping.

2. Distribution in India

The distribution of ornamental fishes varies with river systems, climate, and geography:

a. North-East India : Considered a biodiversity hotspot, Rivers: Brahmaputra and its tributaries, Rich in barbs, danios, loaches, High level of endemism

b. Western Ghats (including Kerala): Another major hotspot with many endemic species, Rivers: Periyar, Chalakudy, Bharathapuzha, Known for colorful barbs and cichlids like Etroplus maculatus

c. Eastern and Central India: Rivers: Ganga, Mahanadi, Godavari, Moderate diversity of cyprinids and catfishes

d. Northern India: Himalayan streams and rivers, Coldwater ornamental species like hill-stream fishes

e. Southern Peninsular rivers: Cauvery, Krishna, and other river systems, Support diverse ornamental fish fauna

3. Factors influencing distribution: Water temperature and flow, Substrate type (sand, rocks, vegetation), Availability of food and shelter, Altitude and rainfall patterns

4. Importance :High export potential in the ornamental fish trade, Supports livelihood of local communities, Important for biodiversity conservation and research

5. Challenges : Overexploitation from wild collection, Habitat destruction and pollution, Introduction of exotic species

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